15.2.2 Exercises - Molecular kinetic energy

1. In a sample of gas at room temperature, five atoms have the following speeds:

1.32×10^3 m s^{-1}
1.50×10^3 m s^{-1}
1.46×10^3 m s^{-1}
1.28×10^3 m s^{-1}
1.64×10^3 m s^{-1}.

For these five atoms, calculate, to three significant figures,

a. the mean speed,

mean speed = ...ms^{-1} [1]

b. the mean-square speed,

Mean-square speed = ...m^2s^{-2} [2]

c. the root-mean-square speed.

root-mean-square speed = ...ms^{-1} [1]

[Total: 4]

2.

a. The kinetic theory of gases is based on some simplifying assumptions.
The molecules of the gas are assumed to behave as hard elastic identical spheres.
State the assumption about ideal gas molecules based on

i. the nature of their movement,

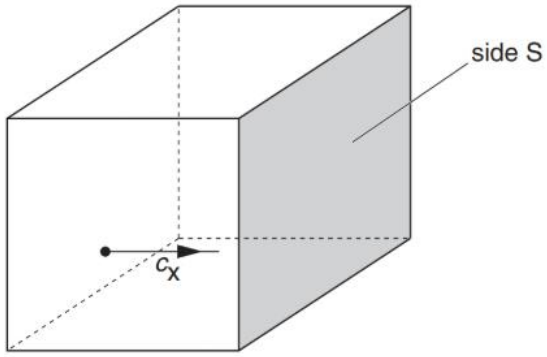
.....[1]

ii. their volume.

.....

.....[1]

b. A cube of volume V contains N molecules of an ideal gas. Each molecule has a component c_X of velocity normal to one side S of the cube, as shown in Fig.



The pressure *p* of the gas due to the component c_X of velocity is given by the expression

$$pV = Nmc_x{}^2$$

Explain how the expression leads to the relation [3]

$$pV = \frac{1}{3}Nm < c^2 >$$

c. The molecules of an ideal gas have a root-mean-square (r.m.s.) speed of 520 m s^{-1} at a temperature of 27 °C.

Calculate the r.m.s. speed of the molecules at a temperature of 100 °C.

root-mean-square speed = ...ms^{-1} [3]

[Total: 8]

3.

a. Explain qualitatively how molecular movement causes the pressure exerted by a gas.

.....

.....

.....[3]

b. The density of neon gas at a temperature of 273 K and a pressure of 1.02×10^5 Pa is 0.900 kg m^{-3} . Neon may be assumed to be an ideal gas.

Calculate the root-mean-square (r.m.s.) speed of neon atoms at

i. 273 K,

speed = ...ms^{-1} [3]

ii. 546 K.

speed = ...ms^{-1} [2]

c. The calculations in (b) are based on the density for neon being 0.900 kg m^{-3}.

Suggest the effect, if any, on the root-mean-square speed of changing the density at constant temperature.

.....

.....[2]

[Total: 10]

4.

a. The equation of state for an ideal gas can be written as pV = NkT.

State the meaning of each of the symbols in this equation.

p: .....

V: .....

N: .....

k: .....

T: .....[3]

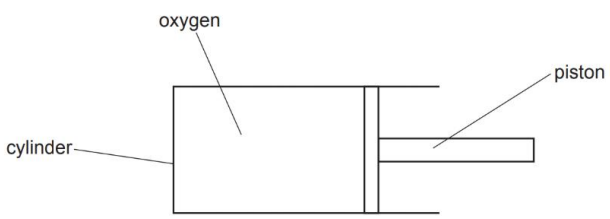
b. Use the equation in (a) to show that the average translational kinetic energy EK of a molecule of an ideal gas is given by $E_K = \frac{3}{2}kT$. [2]

c. The mass of an oxygen molecule is 5.31×10^{-26} kg. Assume that oxygen behaves as an ideal gas.

i. Use the equation in (b) to determine the root-mean-square (r.m.s.) speed u of an oxygen molecule at 23°C.

u = ...ms^{-1} [3]

ii. A fixed mass of oxygen gas at initial pressure P is sealed in a cylindrical container by a movable piston at one end, as shown in Fig.



The temperature of the gas is 23°C.
The piston is slowly moved into the cylinder so that the oxygen gas is compressed. At all times, the gas and the container remain in thermal equilibrium with the surroundings.
On Fig. 3.2, sketch the variation with pressure of the r.m.s. speed of the oxygen molecules as the pressure increases. [2]

[Total: 10]